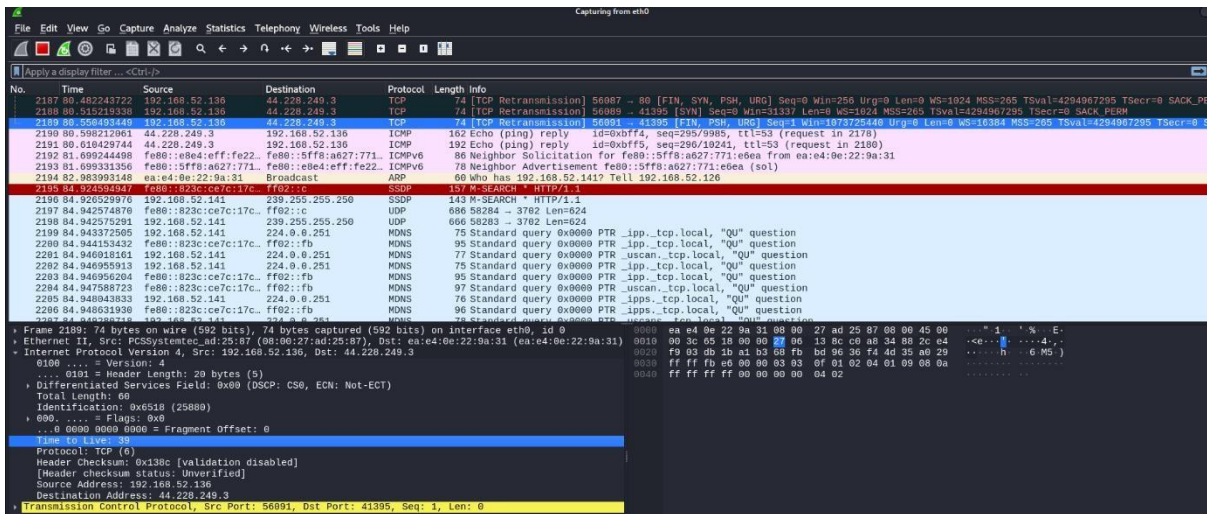


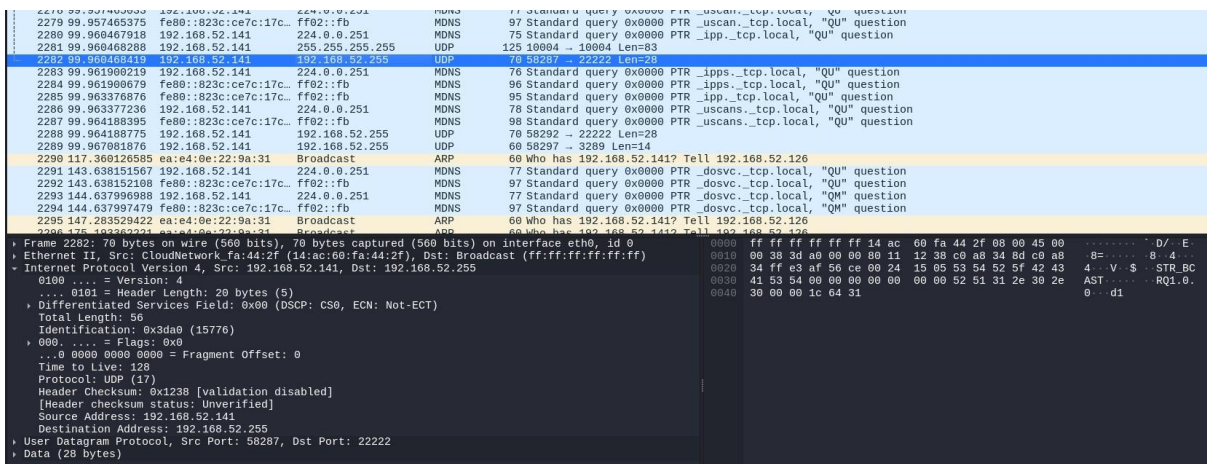
OS Fingerprinting using NMAP

TTL (Time-To-Live) is a field in IP packets that indicates the maximum number of hops a packet can take before being discarded. In Wireshark, you can check TTL values in the IP header to analyze OS fingerprinting (e.g., Windows typically uses 128, Linux 64).

So here I tried for TCP and it showed the TTL of 39..which is basically linux machine!



So the same here I tried for UDP and it showed the TTL of 120..which is basically windows machine!



So I tried the same for ssdp(simple service discovery protocol) and it showed the TTL is of in HOP LIMIT=4..!

```

2250 90.794857415 fe80::823c:ce7c:17c... ff02::c UDP 686 58284 - 3702 Len=624
2251 90.883655288 192.168.52.141 239.255.255.255 UDP 666 58283 - 3702 Len=624
2252 90.928701913 fe80::823c:ce7c:17c... ff02::c SSDP 157 58280 - 1117415
2253 90.928702011 192.168.52.141 239.255.255.255 SSDP 143 M-SEARCH * HTTP/1.1
2254 90.955661294 192.168.52.141 255.255.255.255 UDP 125 10004 - 10004 Len=83
2255 90.956411284 192.168.52.141 192.168.52.255 UDP 70 58287 - 22222 Len=28
2256 90.960129386 192.168.52.141 192.168.52.255 UDP 70 58292 - 22222 Len=28
2257 90.962672246 192.168.52.141 192.168.52.255 UDP 60 58297 - 3289 Len=14
2258 91.950675977 192.168.52.141 224.0.0.251 MDNS 75 Standard query 0x0000 PTR _ipp._tcp.local, "QU" question
2259 91.950677652 fe80::823c:ce7c:17c... ff02::fb MDNS 95 Standard query 0x0000 PTR _ipp._tcp.local, "QU" question
2260 91.951461224 192.168.52.141 224.0.0.251 MDNS 77 Standard query 0x0000 PTR _uscan._tcp.local, "QU" question
2261 91.952144424 fe80::823c:ce7c:17c... ff02::fb MDNS 97 Standard query 0x0000 PTR _uscan._tcp.local, "QU" question
2262 91.952878197 192.168.52.141 224.0.0.251 MDNS 75 Standard query 0x0000 PTR _ipp._tcp.local, "QU" question
2263 91.954664268 fe80::823c:ce7c:17c... ff02::fb MDNS 95 Standard query 0x0000 PTR _ipp._tcp.local, "QU" question
2264 91.954664649 192.168.52.141 224.0.0.251 MDNS 78 Standard query 0x0000 PTR _uscans._tcp.local, "QU" question
2265 91.956608709 192.168.52.141 224.0.0.251 MDNS 76 Standard query 0x0000 PTR _ipps._tcp.local, "QU" question
2266 91.956609899 fe80::823c:ce7c:17c... ff02::fb MDNS 96 Standard query 0x0000 PTR _ipps._tcp.local, "QU" question
2267 91.957375913 fe80::823c:ce7c:17c... ff02::fb MDNS 98 Standard query 0x0000 PTR _uscans._tcp.local, "QU" question
2268 93.957429556 192.168.52.141 255.255.255.255 UDP 125 10004 - 10004 Len=83
2269 93.958918441 192.168.52.141 192.168.52.255 UDP 70 58297 - 22222 Len=28

```

Frame 2252: 157 bytes on wire (1256 bits), 157 bytes captured (1256 bits) on interface eth0, id 0
 Ethernet II, Src: CloudNetwork_fa:44:2f (14:ac:60:fa:44:2f), Dst: IPv6mcast_0c (33:33:00:00:00:0c)
 Internet Protocol Version 6, Src: fe80::823c:ce7c:17c:f163, Dst: ff02::c
 0110 ... = Version: 0
 ... 0000 0000 ... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
 ... 0000 00 ... = Differentiated Services Codepoint: Default (0)
 ... 0000 00 ... = Explicit Congestion Notification: Not ECN-Capable Transport
 ... 1100 0100 1111 0010 0011 = Flow Label: 0xc4f23
 Payload Length: 103
 Next Header: UDP (17)
 Hop Limit: 4
 Source Address: fe80::823c:ce7c:17c:f163
 Destination Address: ff02::c
 User Datagram Protocol, Src Port: 54367, Dst Port: 1900
 Simple Service Discovery Protocol

Cmd : -O for checking OPERATING SYSTEM of the target!

```

root@kali: ~/home/kali
# nmap -O testphp.vulnweb.com
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-27 09:12 EDT
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.051s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
OS fingerprint not ideal because: Missing a closed TCP port so results incomplete
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.93 seconds

```

Cmd : -sV for checking software version information of the target!

```

root@kali: ~/home/kali
# nmap -sV testphp.vulnweb.com
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-27 09:22 EDT
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.050s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http      nginx 1.19.0
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 39.17 seconds

```

Cmd : -sS Uses SYN scan (stealthy, doesn't complete TCP handshake).

```

root@kali: ~/home/kali
# nmap -sS testphp.vulnweb.com
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-27 09:27 EDT
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.050s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
Nmap done: 1 IP address (1 host up) scanned in 17.93 seconds

```

Cmd : -f Breaks packets into tiny 8-byte fragments to evade detection.

```

(root@kali)-[/home/kali]
# nmap -f testphp.vulnweb.com
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-27 09:30 EDT
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.32s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
All 1000 scanned ports on testphp.vulnweb.com (44.228.249.3) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 329.58 seconds

```

Cmd : -mtu 16 Sets Maximum Transmission Unit (MTU) to fragment packets into 16-byte chunks.

```

(root@kali)-[/home/kali]
# nmap -mtu 16 testphp.vulnweb.com
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-27 09:44 EDT
Stats: 0:00:54 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 14.00% done; ETC: 09:50 (0:05:26 remaining)
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.37s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
All 1000 scanned ports on testphp.vulnweb.com (44.228.249.3) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 375.87 seconds

```

Cmd : -D RND:10 Generates 10 random decoy IPs, confusing IPS logs.

```

(root@kali)-[/home/kali]
# nmap -D RND:10 testphp.vulnweb.com
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-27 09:52 EDT
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.056s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 13.05 seconds

```

Cmd : -sA Sends ACK packets, tricking some IDS/IPS into thinking it's established traffic.

```

(root@kali)-[/home/kali]
# nmap -sA 44.228.249.3
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-27 10:03 EDT
Nmap scan report for ec2-44-228-249-3.us-west-2.compute.amazonaws.com (44.228.249.3)
Host is up (0.047s latency).
All 1000 scanned ports on ec2-44-228-249-3.us-west-2.compute.amazonaws.com (44.228.249.3) are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 1.87 seconds

```